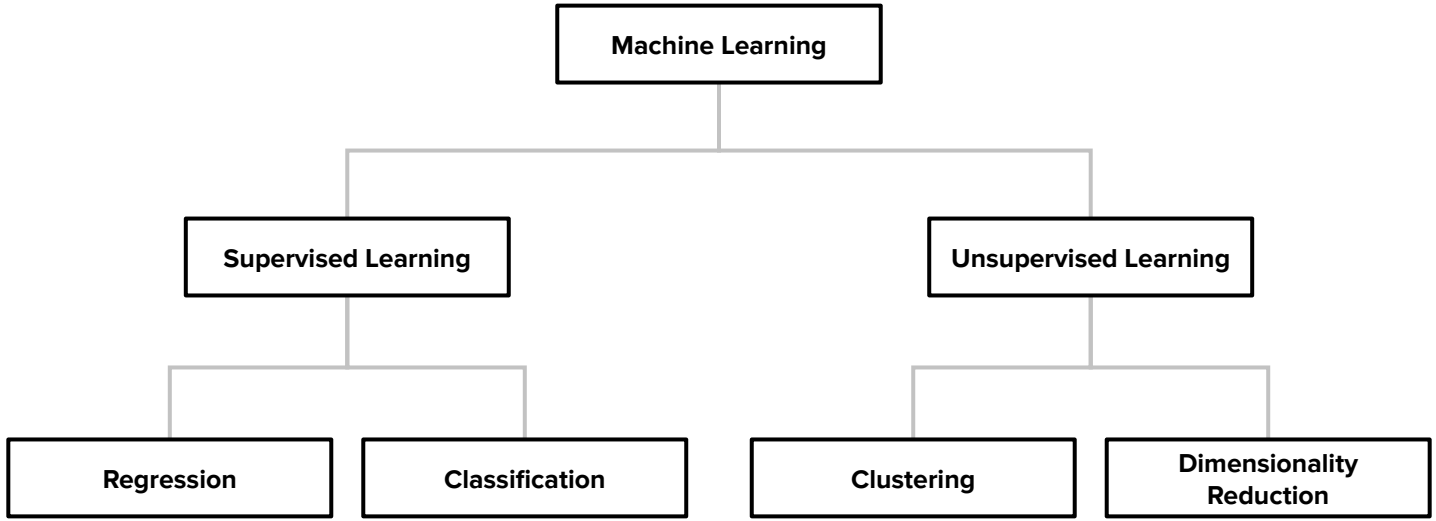
Classification and Logistic Regression

GENERAL ASSEMBLY

Agenda

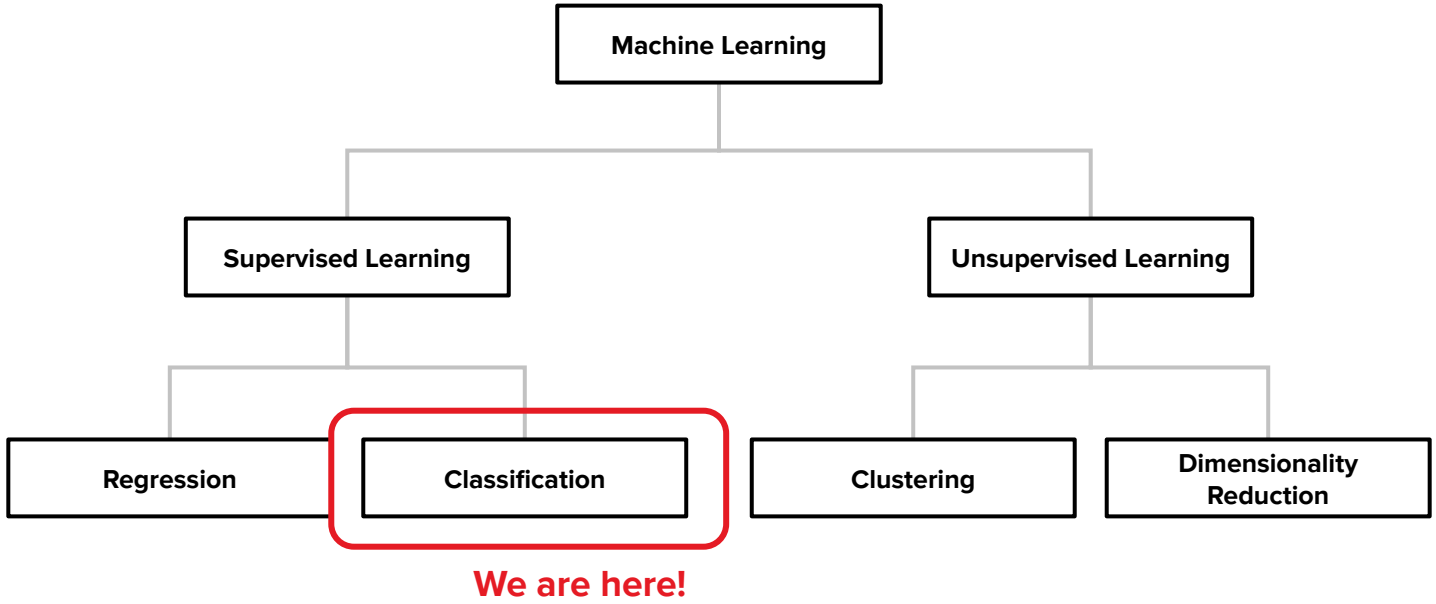
- How does classification differ from regression?
- An overview of how logistic regression work
- Carry out logistic regressions in Python
- Interpret results and make probabilistic predictions

Where are we now?



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Where are we now?



Classification

Classification is the branch of supervised learning in which we are concerned with predicting a **categorical response**, oftentimes a binary (yes/no) response. For example:

“Given this person’s demographics, will they vote for candidate A or B?”

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Classification

Classification is the branch of supervised learning in which we are concerned with predicting a **categorical response**, oftentimes a binary (yes/no) response. For example:

“Will it rain tomorrow?”

Classification

Classification is the branch of supervised learning in which we are concerned with predicting a **categorical response**, oftentimes a binary (yes/no) response. For example:

"Given this person's financial status, will they default on their loan?"

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Logistic Regression

Our goal

Remember, our goal is to predict some binary response variable given some *x*-variable(s). Let's denote the probability of this occurring by *p*. Just like linear regression, logistic regression is a **linear model**. That is, we can write it out as:

$$\text{something} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

But what is "something"?

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Attempt #1

What if we try this?

$$p = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

However, we face a problem when our prediction is outside the interval 0 and 1.

Attempt #2

$$\frac{p}{1-p} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

This might look weird, but that thing on the left can now take on values between 0 and infinity. So this is better! But it still can't handle negative values.

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Aside: Odds

The **odds** of an event occurring is:

$$\frac{\text{prob. of the event occurring}}{\text{prob. it does not occur}} = \frac{p}{1-p}$$

Aside: Odds

For example, what are the odds of rolling 1 on a fair 6-sided die?

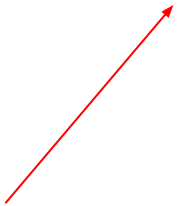
$$\text{odds} = \frac{1/5}{5/6} = \frac{1}{5} = 5:1$$

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Aside: Odds

For example, what are the odds of rolling 1 on a fair 6-sided die?

$$\text{odds} = \frac{1/5}{5/6} = \frac{1}{5} = 5:1$$



Gamblers prefer a colon with the bigger number first because it represents their payoff

Attempt #3

$$\log \frac{p}{1-p} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

It may be hard to see, but that actually does it! Adding a natural log on the left, it can now take on any number the right side might accidentally throw at it. This is the general form of **logistic regression**!

Where does this get us?

Using some algebra, we can actually rearrange

$$\log \frac{p}{1-p} = \beta_0 + \beta_1 x$$

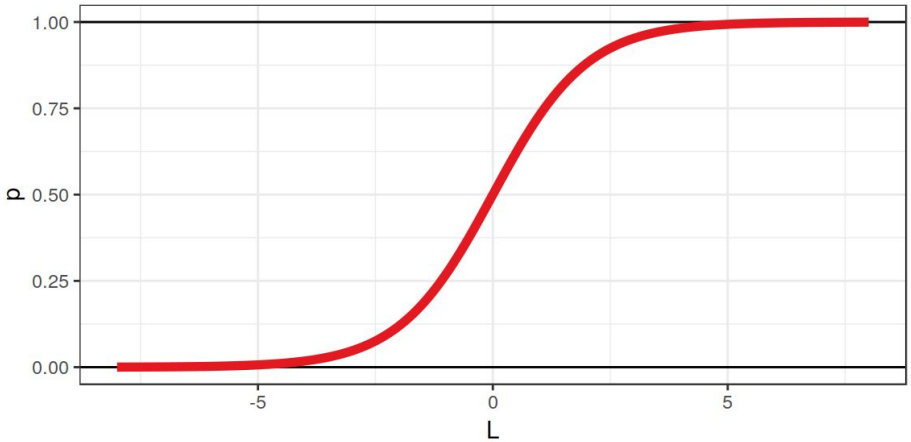
into

$$p = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}}$$

Great!

So, given our estimates for β_0 and β_1, we can then compute the probability of a given event occurring based on some *x*!

$$p = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}} = \frac{e^{L}}{1+e^{L}}$$



Let’s do some ourselves!



Exercise: Interpreting Slope

Let's recap how we interpret the slope of linear regression:

$$\begin{aligned} y &= \beta_0 + \beta_1(x+1) \\ &= \beta_0 + \beta_1 x + \beta_1 \\ &= (\beta_0 + \beta_1 x) + \beta_1 \end{aligned}$$

"For a one-unit increase in x, we expect y to increase by β_1."

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Exercise: Interpreting Slope

To do this for logistic regression, we'll need to exponentiate both sides:

$$\frac{p}{1-p} = \text{odds} = e^{\beta_0 + \beta_1 x}$$

Exercise: Interpreting Slope

Now let's see how the odds are affected when we increase *x* by 1.

$$\begin{aligned} \text{odds} &= e^{\beta_0+\beta_1(x+1)} \\ &= e^{\beta_0+\beta_1 x+\beta_1} \\ &= e^{\beta_0+\beta_1 x}\, e^{\beta_1} \end{aligned}$$

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Exercise: Interpreting Slope

Now let's see how the odds are affected when we increase *x* by 1.

$$\begin{aligned} \text{odds} &= e^{\beta_0+\beta_1(x+1)} \\ &= e^{\beta_0+\beta_1 x+\beta_1} \\ &= e^{\beta_0+\beta_1 x} e^{\beta_1} \end{aligned}$$

The exponentiation of β_1 is now a multiplicative factor of odds!

Exercise: Interpreting Slope

Now let's see how the odds are affected when we increase *x* by 1.

$$\begin{aligned} \text{odds} &= e^{\beta_0+\beta_1(x+1)} \\ &= e^{\beta_0+\beta_1 x+\beta_1} \\ &= e^{\beta_0+\beta_1 x} e^{\beta_1} \end{aligned}$$

For a one-unit increase in x, we expect the odds of success to increase by a factor of exp(β_1).

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Back to the code!

